MIL-I-38133A(USAF)
14 November 1967
Superseding
MIL-I-38133 (USAF)
5 April 1963

MILITARY SPECIFICATION

INDICATOR, PRESSURE, OIL

1. SCOPE

1.1 This specification covers hermetically sealed, remote indicating, voltage ratio type, integrally lighted, oil pressure indicators.

1.2 Classification. The indicators shall be of the following types, as specified (see 6.2):

Type I Indicator, Pressure, Oil EGU-10/A - White lighting

Type II Indicator, Pressure, Oil EGU-11/A - Red lighting

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-675	Coating of Glass Optical Elements (Anti-reflection)
Mil-I-25437	Indicators, Aircraft, Pressure, Voltage Ratio Type, General Specification for
Mil-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-C-26500	Connector, General Purpose, Electrical Miniature, Circular, Environment Resisting, 200°C Ambient Temperature
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-T-38230	Transmitter, Pressure TRU-66/A, Oil, Variable Reluctance

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-781	Reliability Tests Exponential Distribution

FSC 6685

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MIL-STD-785	Requirements for Reliability Program (for Systems and Equipments)
MS27034	Connector, Receptacle, Electrical, Pin Insert, Cylindrical, Miniature Hermetic, Solder Mount
MS33639	Case, Instrument, Clamp-Mounted, Aircraft

(Copies of documents required by suppliers in connection with specific procurement functions may be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 General specification. The requirements of MIL-I-25437 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Design and construction

3.3.1 The indicator shall be designed for use with the TRU-66/A transmitter conforming to MIL-T-38230.

3.3.2 Case. The case dimensions shall conform to MS33639 for the 1-inch size. The case length shall not exceed 4.0 inches.

3.3.3 Electrical connector. The indicator shall be provided with an MS27034H10T5P electrical connector conforming to MIL-C-26500.

3.3.4 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-785.

3.3.4.1 Reliability. The indicator shall have a minimum mean-time-between-failures (MTBF) of 2,000 hours at a confidence level of 90 percent and a discrimination ratio of 1.5.

3.3.4.2 Longevity. The indicator shall have an operating life span of 3,000 hours.

3.4 Performance

3.4.1 Tolerances. The indicator shall operate within the tolerances specified in tables I and II.

3.5 Lighting. The indicators shall be integrally lighted and shall meet all requirements specified herein. The type I indicator (EGU-10/A) shall be lighted in accordance with MIL-L-27160, and the type II indicator (EGU-11/A) shall be lighted in accordance with MIL-L-25467.

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TABLE I. Temperature Scale Error

Test Pressure PSI	Scale Error	
	Room Temperature PSI	High and Low Temp Exposure High and Low Temperature PSI
0	1.50	3.00
10	1.50	3.00
20	1.50	3.00
30	1.50	3.00
40	1.50	3.00
50	1.50	3.00
60	1.50	3.00
70	1.50	3.00
80	1.50	3.00
90	1.50	3.00
100	1.50	3.00

TABLE II. Position, Friction, and Vibration Error

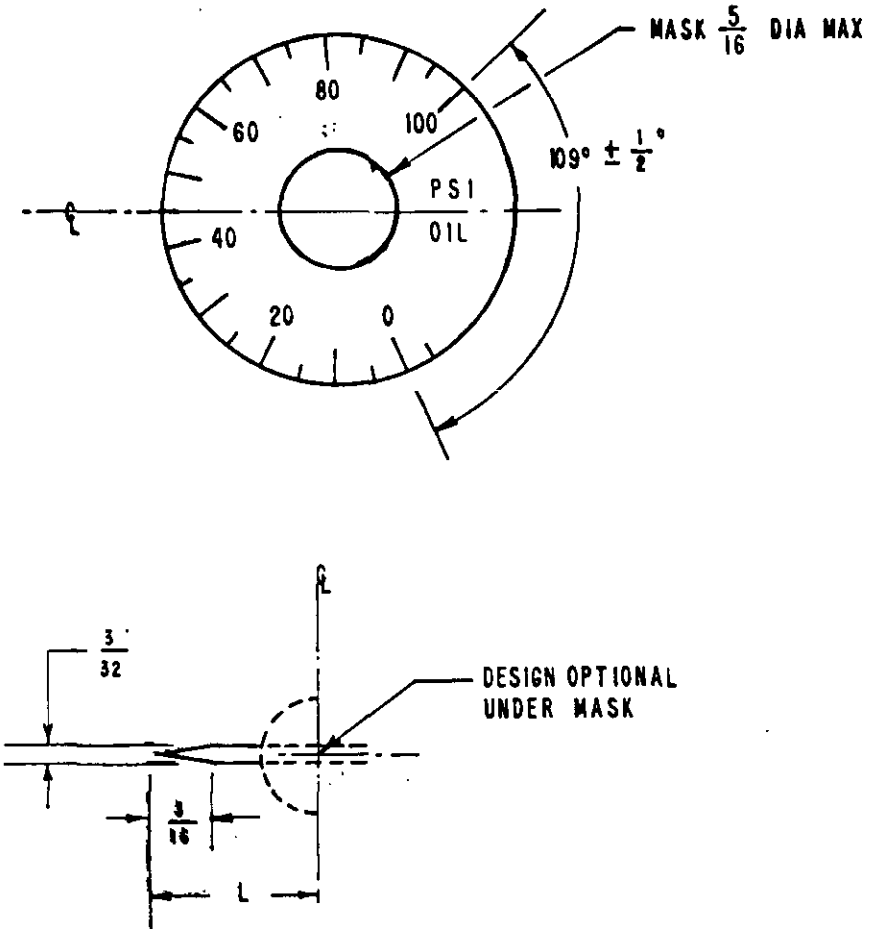
Test	Tolerance ± PSI
Position error at 50 psi	1.5
Friction	
Room temperature	1.5
High temperature	2.00
Low temperature	2.25
Vibration	
Pointer oscillation	2.0
Pointer variation	1.5

3.6 Pointer. The pointer shall conform to figure 1 and shall be suitable for integral lighting. The length of the pointer shall be such that the pointer tip will extend into the scale a distance equal to approximately one-third the length of the shortest graduation.

3.6.1 Finish. The lighted portion of the pointer shall be finished in color No. 37875 of FED-STD-595.

3.7 Dial. The dial display shall conform to figure 1. The 45-psi graduation shall be located on the horizontal centerline at the left side of the dial.

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DIMENSIONS IN INCHES

UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ± 1/64

FIGURE 1. Indicator Dial

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3.7.1 Markings

3.7.1.1 White. The following markings shall be finished in durable white, color No. 37875 of FED-STD-595. The dimensions shall be as specified.

	Height or Length Inch +1/64	Width of Line or Graduation Inch ±0.005
Numerals 0.20.40, 60, 80, and 100	3/32	0.0213
10-Pound graduations	7/64	0.030
5-Pound graduations	5/64	0.015
Lettering OIL	5/64	0.015
Lettering PSI	5/64	0.015

3.7.1.2 Black. The markings USAF and EGU-10/A or EGU-11/A, as applicable, shall be permanently and legibly marked on the side of the indicator in 1/8-inch-high letters. Unless otherwise specified, these and all other markings shall be finished in durable black, color No. 37038 of FED-STD-595.

3.8 Electrical wiring. Internal and external wiring shall conform to figure 2. The internal wiring shall be color coded and connected to the receptable as shown.

3.9 Electrical characteristics. The current used by the indicator shall not exceed 75 ma. The power consumption, exclusive of the lamps, shall not exceed 0.60 watt.

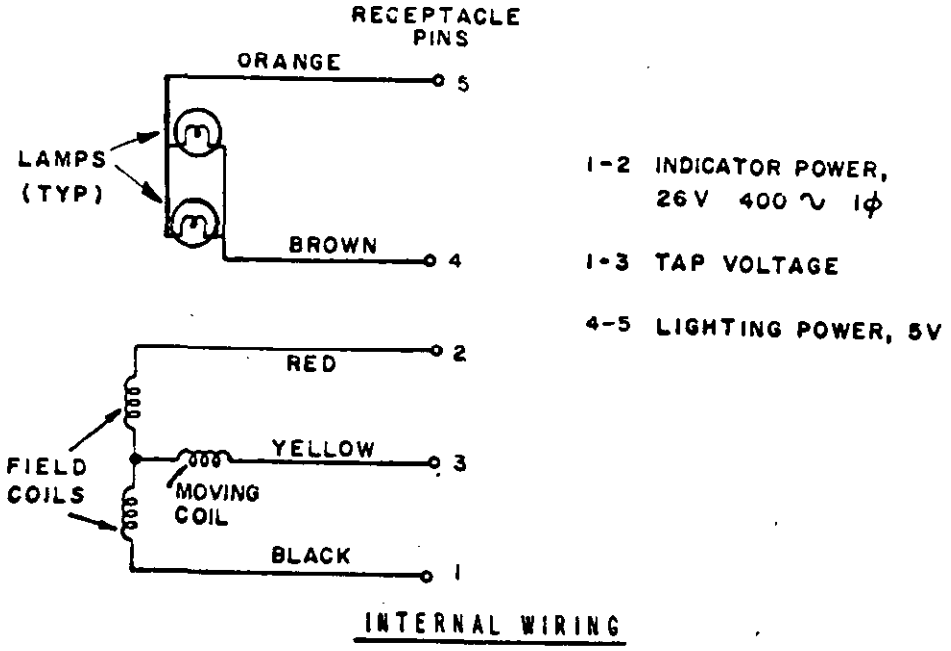
3.10 Percent reflectance. All reflecting glass surface shall be provided with a reflectance reducing coating which meets the requirements of MIL-C-675, except for the materials used, in addition to withstanding the environmental conditions specified herein. The following reflectance tolerances shall apply:

Angle of Incidence	Wavelength Millimicrons	Percent Reflectance
0°	450 - 675	0.6 absolute
0°	425 - 700	0.5 average
30°	450 - 625	1.0 absolute
30°	425 - 700	0.5 average.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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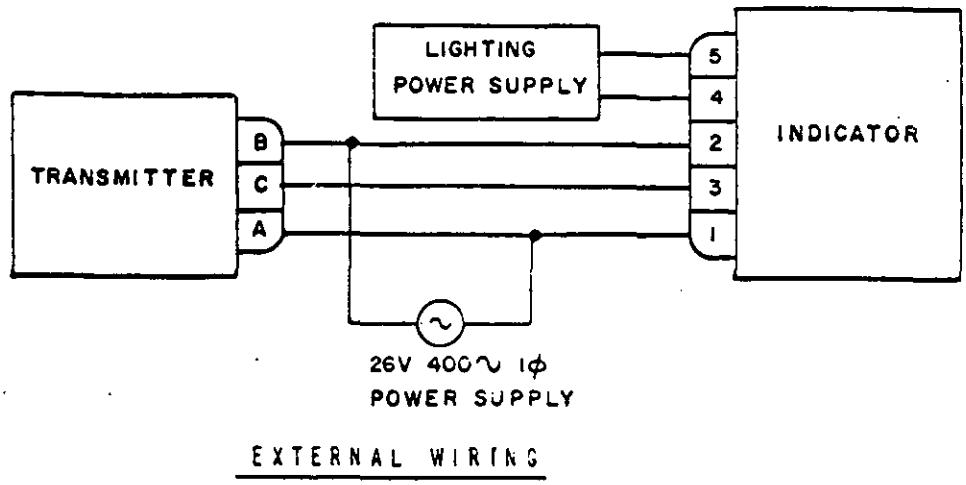


FIGURE 2. Wiring Diagram

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4.2 General specification. The inspection and testing of indicators shall be in accordance with MIL-I-25437 and as specified herein.

4.3 Qualification testing

4.3.1 Test samples. The test samples shall consist of 10 indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity. Seven of the samples shall be subjected to the reliability test only and the other three shall be subjected to all the tests except reliability.

4.3.2 Test report. The contractor shall prepare and submit a test report in accordance with MIL-STD-831.

4.3.3 Qualification tests. The qualification tests shall consist of all the tests specified herein and in MIL-I-25437.

4.3.3.1 Nonstandard parts data approval. When nonstandard parts are used in the construction of the indicator, nonstandard parts data approval shall be a prerequisite to qualification approval.

4.4 Test methods

4.4.1 Percent reflectance. The reflectance of all glass surfaces shall be measured and the tolerances shall not exceed those specified in 3.10.

4.4.2 Lighting. The type I indicator shall be subjected to lighting tests in accordance with MIL-L-27160 and the type II indicator shall be subjected to lighting tests in accordance with MIL-L-25467.

4.4.3 Reliability. Seven indicators shall be tested for reliability. Acceptance permitting production and total test time shall be in accordance with MIL-STD-781, test plan I. Recording, data handling, and reporting shall also be in accordance with MIL-STD-781.

4.4.3.1 Test. The test equipment and fixtures shall include provisions for vibration and temperature cycling at the periodic intervals specified herein. During this test, input signals shall be applied to the indicator at a rate of 2 to 3 cycles (minimum and maximum) per minute from 0 to 85 percent of full scale and return. The test shall begin when rated power is applied. After completion of 4 hours of operation with rated power on, which shall include at least 480 cycles of operation, the power shall be turned off and shall remain off for a period of 10 minutes. For at least 10 minutes out of every hour of ON time, the indicator shall be vibrated in accordance with test level J of MIL-STD-781. The temperature cycles shall be conducted in the following sequence: 4 hours at -20°C, 14 minutes for temperature change to +50°C at a minimum rate of 5°C per minute: 4 hours at +50°C, 14 minutes for temperature change to -20°C, and 4 hours at -20°C. In case of malfunction, the malfunction shall be recorded and the test

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interrupted while repairs are made. Upon completion of each 250 hours of cycling, the indicator shall be checked for scale error. The scale error reading shall not exceed 150 percent of the room temperature tolerance in table II. No recalibration shall be performed on the units during this test. The test may be interrupted for weekend shutdown. The contractor shall prepare a report in accordance with MIL-STD-781 which shall contain test procedures, details of any malfunctions, and corrective actions taken.

4.4.3.2 Longevity. At the completion of the reliability test, testing of three of the indicators shall be continued for a total of 3,000 hours each. The test time accumulated during the reliability test shall be applied to this test. The accept-reject criteria is not applicable.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-I-25437 and as follows:

5.1.1 Marking for shipment. The shipment marking nomenclature shall be as follows, as applicable:

Indicator, Pressure, Oil EGU-10/A
or
Indicator, Pressure, Oil EGU-11/A.

6. NOTES

6.1 Intended use. The EGU-10/A and EGU-11/A indicators covered by this specification are intended for use with the TRU-66/A transmitter in indicating remotely the oil pressure of gas turbine engines where panel space is of prime importance.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type of indicator required (see 1.2)

c. When sampling plan B tests will not be conducted

d. Levels of packaging and packing required.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to

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the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Systems Engineering Group, Attn: SE FIP, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 11

Review activity:
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